Mechanical Engineering Experiments (II)-Worksheet for Project Meeting

機械工程實驗(下)-專題討論工作單

Group information: (Fill your group number here No. 5)

Group Leader (Name and Student ID) : 李承軒 B12502027
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Read the following notes before editing
1. DO NOT extend the space provided. Learn to select representative points and leave detailed explanations to your slides. You may adjust the layout within the designated page if needed.
2. Just leave blank for those slots you are not going to fill.

Meeting date:

Credit:

Group leader's signature:

Feedback from Lecturer:

Lecturer signature:

Part (a) What have you done

1. **Subsystem Design and Fabrication:** We designed and built a new three-ball collecting subsystem for the drone. The new mechanism keeps the same 8-screw mounting method as the midterm design, so the original drone structure does not need to be changed. We also changed the system from a three-servo design to a five-servo design to improve the stability of collecting three balls at once. In addition, we evaluated the center of gravity and reorganized the wiring with a 9×2 Dupont header to make the subsystem cleaner and more stable.
2. **Safety Evaluation:** We estimated the flight of a full battery. Theoretically, the effectiveness of each device is quite similar. They can both fly for around 16 minutes if the battery is fully charged, though 3-ball one can fly seconds longer. We also measured the area of the bottom of the devices in order to evaluate the possibility of landing on the plate.

3. **Final Strategy Planning:** Since we all agreed that our goal for the final test is to obtain 100 points, we created a strategy table. The final test provides four attempts, and there is also a retest mechanism that can be triggered if we receive two zero scores. This table allows us to quickly determine the best strategy for the remaining attempts after knowing the scores from the previous attempts, ensuring that we can secure 100 points in the final test as reliably as possible.

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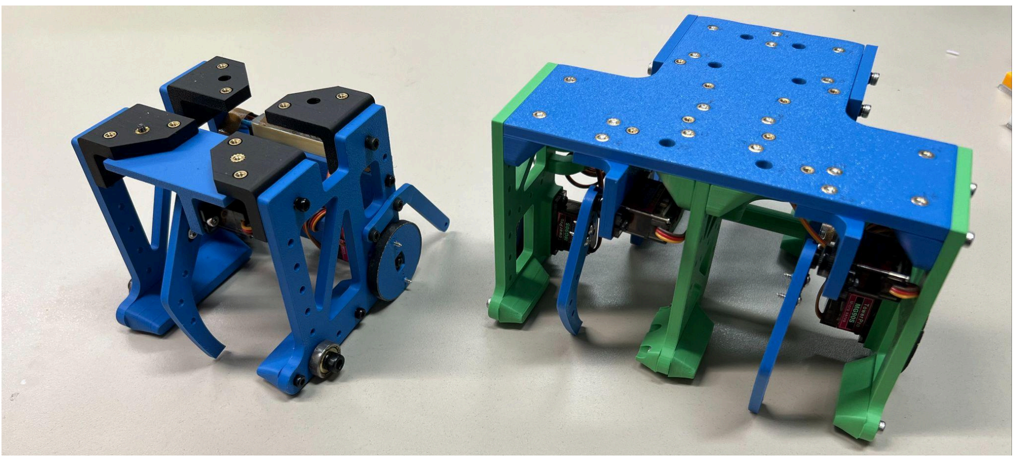

Part (b) What barriers you encounter:

1. **Analysis goals unable to achieve**:
   Our initial goal was to link drone position above the pad to landing drift. Rough hover tests using a rope showed no clear correlation, suggesting drift is likely a stochastic issue. However, observed variations in aerial behavior at different altitudes offer a new potential area for investigation.

2. **The new mechanism didn't reach our expectation:**
   Contrary to expectations, the new mechanism's completion time improved only marginally. A multi-aspect comparison between the tri-ball and single-ball systems follows.:
   (1) Aerial Maneuverability:
   a. **Mission Completion Time:**
   With both mechanism tested for 3 times:
   ○ The **Single-Ball Mechanism (**Total time: 67.9s.**):**
   Air time (A to B or B to A): 7.3s, Air time(Start to A): 5s
   Ground time (per ball): 8.8s, Max Ground time: 11s, Min Ground time: 5s.
   ○ The **Tri-Ball Mechanism (**Total time: 56.8s.**):**
   Air time (A to B or B to A): 10s, Air time(Start to A): 9s
   Ground time (per ball): 12.6, Max Ground time: 25s, Min Ground time: 6s.
   Tri-ball ground durations are significantly higher because retrieving balls from difficult corners is time-consuming, unlike the consistent durations of the single-ball unit.

   b. **Flight Handling (Stability during transport):**
   A significant observation noted by the pilot is that the newly developed mechanism exhibits increased instability during platform landings. While the system remains operational, a comprehensive analytical investigation of this behavior will be conducted for the final report.

   (2) Ball-Picking (Ground Phase):
   a. **Limited mobility in Box A:**The three-ball machine has a larger base, making it harder to move, align, and collect balls in the limited space of Box A, especially for balls blown into the corners.
   b. **Balls inside affect control:** When balls are already inside the vehicle, their friction with the floor and mechanism makes straight movement and turning less smooth, increasing the time needed to collect more balls.
   c. **Unintuitive bidirectional control:** Since the vehicle has two front gates and one rear gate, using the rear gate makes the controls reversed, which is difficult and unintuitive to operate.
   d. **Conclusion:** Although the three-ball machine could be improved through practice, there is not enough time before the final test. Therefore, we prefer to use the more stable and easier-to-control single-ball machine.

Part (c) What is your next step:

1. **Improvement of maneuverability with ball transfer unit**

   We plan to replace the current ball bearing support with a ball transfer unit to improve the maneuverability of the ground-vehicle subsystem on the platform. Compared with the original ball bearing support, the ball transfer unit allows smoother multidirectional motion and reduces resistance during turning or small position adjustments. This modification is expected to help the vehicle move more smoothly, approach the target balls more accurately, and improve the overall efficiency of the ball-collection process.

2. **Laser Indicator for Landing:**

   To improve landing accuracy, we plan to add a visual guidance system to help the pilot identify the landing area more clearly. During operation, the pilot must wear a full-face shield for safety, but the shield may distort the field of view and make it more difficult to judge the drone’s position relative to the landing platform. This issue may reduce landing precision, especially because the drone must land accurately before the ground-vehicle subsystem begins the ball-collection task. Therefore, we are considering installing laser indicators as visual references for the landing zone. Depending on the available space on the system and the overall cost limitation, the indicators may be installed either on both sides of the platform or at the four corners. A four-corner setup would provide a clearer boundary of the landing area, while a two-side setup would be simpler, more cost-effective, and easier to integrate into the current structure. As an alternative or backup solution, a rope or other physical marker may also be used to define the landing zone. This improvement is expected to reduce pilot uncertainty, improve landing consistency, and make the transition from drone landing to ball collection smoother and more reliable.

Peer assessment table (Only Copy-paste the member's name and ID, leave contribution blank):

Fill your group number here: No.

Member Name	**Member student ID**	**Contribution %**
李承軒	B12502027	
林岳諦	B12502078	
楊振國	B12502162	
李汪懋	B12502017	
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